

Ground-dwelling beetle communities in newly established suburban yards: the role of residential landscape design

Urban Ecosystems

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Table S1: Beetle species identified across the two neighborhoods, Del Webb and Weslyn Park. Native status and functional traits are reported. Functional traits include diet, wings morphology, body size, and egg-larval substrate. Diet categories are OMN = omnivorous, PRD = predatory, PHY = phytophagy, MYC = mycophagy, DET = detritivory, and TRO = trophallaxis. Wing morphology is classified as BRA = brachypterous and MAC = macropterous. Body size classes include S = small, M = medium, and L = large. Egg-larval substrate categories include SOI = soil, DOM = decayed organic matter, LPL = live plant, and FNG = fungus.

Family	Subfamily	Genus	Species	Author	Diet	Wings	Size (mm)	Size	Substrate	Native Status
Anthicidae	Anthicinae	<i>Acanthinus</i>	<i>argentinus</i>	(Pic, 1913)	OMN	MAC	2.9	S	DOM	N
Anthicidae	Anthicinae	<i>Acanthinus</i>	<i>scitulus</i>	(LeConte, 1852)	OMN	MAC	2.45	S	DOM	Y
Anthicidae	Anthicinae	<i>Anthicus</i>	<i>ictericus</i>	LaFerté-Senectère, 1849	OMN	MAC	1.95	S	DOM	Y
Anthicidae	Anthicinae	<i>Vacusus</i>	<i>parvus</i>	(Pic, 1910)	OMN	MAC	1.9	S	DOM	N
Anthicidae	Tomoderinae	<i>Tomoderus</i>	<i>interruptus</i>	La Ferté, 1848	OMN	MAC	3	S	DOM	Y
Cantharidae	Chauliognathinae	<i>Chauliognathus</i>	<i>marginatus</i>	(Fabricius, 1775)	OMN	MAC	11.5	M	SOI	Y
Carabidae	Scaritinae	<i>Pasimachus</i>	<i>sublaevis</i>	LeConte, 1874	PRD	BRA	25.5	L	SOI	Y
Carabidae	Trechinae	<i>Tachyura</i>	<i>granaria</i>	(Dejean, 1831)	PRD	BRA	2.1	S	SOI	Y
Carabidae	Harpalinae	<i>Tetragonoderus</i>	<i>laevigatus</i>	Chaudoir, 1876	OMN	MAC	4	S	SOI	N
Chrysomelidae	Cryptocephalinae	<i>Pachybrachis</i>	<i>hepaticus</i>	(F. E. Melsheimer, 1847)	PHY	MAC	2.5	S	LPL	Y
Chrysomelidae	Galerucinae	<i>Longitarsus</i>	sp.A		PHY	MAC	1.5	S	LPL	Y
Ciidae	Ciinae	<i>Ciinae</i>	sp.A		MYC	MAC	1.1	S	FNG	Y
Coccinellidae	Scymninae	<i>Diomus</i>	<i>roseicollis</i>	(Mulsant, 1853)	PRD	MAC	1.48	S	LPL	N
Coccinellidae	Scymninae	<i>Scymnus</i>	sp.A		PRD	MAC	1.1	S	LPL	Y
Corylophidae		<i>Corylophidae</i>	sp.A		MYC	BRA	1	S	DOM	Y
Curculionidae	Entiminae	<i>Eudiagogus</i>	<i>rosenschoeldi</i>	Fåhraeus, 1840	PHY	MAC	5.6	S	LPL	Y
Curculionidae	Scolytinae	<i>Scolytinae</i>	sp.A		MYC	BRA	2.2	S	DOM	Y
Curculionidae	Scolytinae	<i>Scolytinae</i>	sp.B		MYC	BRA	2.2	S	DOM	Y
Curculionidae	Dryophthorinae	<i>Sphenophorus</i>	<i>venatus</i>	(Say, 1831)	PHY	MAC	9.5	S	LPL	Y
Dermestidae	Megatominae	<i>Trogoderma</i>	<i>ornatum</i>	(Say, 1825)	DET	MAC	3	S	DOM	Y
Elateridae	Agrypninae	<i>Heteroderes</i>	<i>amplicollis</i>	(Gyllenhal, 1817)	OMN	MAC	8.5	S	SOI	N
Elateridae	Agrypninae	<i>Rismethus</i>	<i>squamiger</i>	(Champion, 1894)	OMN	MAC	2.25	S	SOI	Y
Latridiidae	Corticariinae	<i>Melanophthalma</i>	<i>picta</i>	(LeConte, 1855)	MYC	MAC	1.25	S	FNG	Y
Leiodidae		<i>Leiodidae</i>	sp.A		DET	MAC	1.2	S	DOM	Y
Mycetophagidae		<i>Typhaea</i>	<i>stercorea</i>	(Linnaeus, 1758)	MYC	MAC	2.6	S	DOM	N
Nitidulidae	Nitidulinae	<i>Stelidota</i>	<i>coenosa</i>	Erichson, 1843	DET	MAC	2.25	S	DOM	Y
Noteridae	Noterinae	<i>Hydrocanthus</i>	<i>oblongus</i>	Sharp, 1882	PRD	MAC	4	S	SOI	Y

Family	Subfamily	Genus	Species	Author	Diet	Wings	Size (mm)	Size	Substrate	Native Status
Phalacridae		<i>Stilbusgroup</i>	sp.A		MYC	MAC	1.5	S	DOM	Y
Ptilliidae		<i>Ptillidae</i>	sp.A		MYC	BRA	0.5	S	DOM	Y
Ptilodactylidae	Ptilodactylinae	<i>Ptilodactyla</i>	sp.A		MYC	MAC	5	S	DOM	Y
Ptinidae	Dorcatominae	<i>Caenocara</i>	sp.A		MYC	MAC	1.5	S	FNG	Y
Scarabaeidae	Aphodiinae	<i>Leiopsammodius</i>	malkini	(Cartwright, 1946)	DET	MAC	3.05	S	DOM	Y
Scarabaeidae	Aphodiinae	<i>Platytomus</i>	<i>micros</i>	(Bates, 1887)	DET	MAC	2.5	S	DOM	Y
Scarabaeidae	Cetoniinae	<i>Euphoria</i>	<i>sepulcralis</i>	(Fabricius, 1801)	PHY	MAC	11.35	M	SOI	Y
Scarabaeidae	Cetoniinae	<i>Trigonopeltastes</i>	<i>delta</i>	(Forster, 1771)	PHY	MAC	9.5	S	DOM	Y
Scarabaeidae	Dynastinae	<i>Cyclocephala</i>	<i>lurida</i>	Bland, 1863	PHY	MAC	12	M	SOI	Y
Scarabaeidae	Rutelinae	<i>Anomala</i>	<i>flavipennis</i>	Burmeister, 1844	PHY	MAC	10.55	M	SOI	Y
Silvanidae	Silvaninae	<i>Ahasverus</i>	<i>rectus</i>	LeConte, 1854	MYC	MAC	2	S	LPL	Y
Smicripidae		<i>Smicrips</i>	<i>palmicola</i>	LeConte, 1878	PHY	MAC	1.2	S	DOM	Y
Staphylinidae	Aleocharinae	<i>Aleocharinae</i>	sp.A		PRD	MAC	1	S	DOM	Y
Staphylinidae	Aleocharinae	<i>Aleocharinae</i>	sp.B		PRD	MAC	1.8	S	DOM	Y
Staphylinidae	Aleocharinae	<i>Aleocharinae</i>	sp.C		PRD	MAC	1.5	S	DOM	Y
Staphylinidae	Aleocharinae	<i>Aleocharinae</i>	sp.D		PRD	MAC	1.2	S	DOM	Y
Staphylinidae	Aleocharinae	<i>Aleocharinae</i>	sp.E		PRD	MAC	3	S	DOM	Y
Staphylinidae	Aleocharinae	<i>Aleocharinae</i>	sp.F		PRD	MAC	0.8	S	DOM	Y
Staphylinidae	Aleocharinae	<i>Aleocharinae</i>	sp.G		PRD	MAC	1.8	S	DOM	Y
Staphylinidae	Aleocharinae	<i>Aleocharinae</i>	sp.H		PRD	MAC	3	S	DOM	Y
Staphylinidae	Aleocharinae	<i>Aleocharinae</i>	sp.I		PRD	MAC	2	S	DOM	Y
Staphylinidae	Aleocharinae	<i>Apalonia</i>	<i>seticornis</i>	Casey, 1906	MYC	MAC	1.12	S	SOI	Y
Staphylinidae	Mycetoporinae	<i>Bryoporus</i>	<i>testaceus</i>	LeConte, 1863	MYC	MAC	3.5	S	DOM	Y
Staphylinidae	Mycetoporinae	<i>Ischnosoma</i>	<i>flavicolle</i>	(LeConte 1863)	MYC	MAC	4.25	S	DOM	Y
Staphylinidae	Osoriinae	<i>Mimogonus</i>	<i>fumator</i>	(Fauvel, 1889)	MYC	BRA	2.5	S	DOM	N
Staphylinidae	Osoriinae	<i>Molosoma</i>	<i>brevicorne</i>	(Notman, 1920)	MYC	BRA	4.5	S	DOM	Y
Staphylinidae	Oxytelinae	<i>Apocellus</i>	<i>sphaericollis</i>	(Say, 1834)	DET	MAC	2.5	S	DOM	Y
Staphylinidae	Paederinae	<i>Astenus</i>	sp.A		PRD	MAC	4	S	DOM	Y
Staphylinidae	Paederinae	<i>Astenus</i>	sp.B		PRD	MAC	3.2	S	DOM	Y
Staphylinidae	Paederinae	<i>Hypomedon</i>	<i>debilicornis</i>	(Wollaston, 1857)	DET	MAC	2.5	S	DOM	Y
Staphylinidae	Paederinae	<i>Myrmecosaurus</i>	<i>ferrugineus</i>	Bruch	TRO	MAC	3	S	SOI	N
Staphylinidae	Paederinae	<i>Scopaeus</i>	sp.A		PRD	MAC	3.5	S	DOM	Y

Family	Subfamily	Genus	Species	Author	Diet	Wings	Size (mm)	Size	Substrate	Native Status
Staphylinidae	Scaphidiinae	<i>Scaphisoma</i>	sp.A		MYC	MAC	1	S	FNG	Y
Staphylinidae	Scydmaeninae	<i>Euconnus</i>	sp.A		PRD	MAC	1	S	DOM	Y
Staphylinidae	Staphylininae	<i>Erichsonius</i>	<i>floridanus</i>	Frank, 1975	PRD	MAC	5	S	SOI	Y
Staphylinidae	Staphylininae	<i>Philonthus</i>	sp.A		PRD	MAC	4	S	DOM	Y
Staphylinidae	Staphylininae	<i>Platydracus</i>	<i>caliginosus</i>	(Erichson, 1839)	PRD	MAC	14.5	M	DOM	Y
Staphylinidae	Xantholininae	<i>Diochus</i>	<i>schaumii</i>	Kraatz, 1860	PRD	BRA	3.8	S	DOM	Y
Staphylinidae	Xantholininae	<i>Lithocharodes</i>	<i>ruficollis</i>	(LeConte, 1863)	PRD	MAC	4.3	S	DOM	Y
Staphylinidae	Xantholininae	<i>Xantholininae</i>	sp.A		PRD	BRA	3.2	S	DOM	Y
Staphylinidae	Xantholininae	<i>Xantholininae</i>	sp.B		PRD	BRA	3	S	DOM	Y
Staphylinidae	Xantholininae	<i>Xantholininae</i>	sp.C		PRD	BRA	4	S	DOM	Y
Tenebrionidae	Alleculinae	<i>Lobopoda</i>	sp.A		DET	MAC	10	M	DOM	Y
Tenebrionidae	Blaptinae	<i>Asiopus</i>	<i>minimus</i>	(Palisot de Beauvois, 1817)	DET	BRA	9	S	SOI	Y
Tenebrionidae	Blaptinae	<i>Blapstinus</i>	<i>fortis</i>	LeConte, 1878	DET	MAC	8.1	S	SOI	Y
Tenebrionidae	Diaperinae	<i>Gondwanocrypticus</i>	<i>platensis</i>	(Fairmaire, 1884)	DET	MAC	4.25	S	DOM	N
Tenebrionidae	Diaperinae	<i>Poecilocrypticus</i>	<i>formicophilus</i>	Gebien, 1928	DET	MAC	2	S	DOM	N

Table S2: Plant species identified across the two neighborhoods, Del Webb and Weslyn Park. Native status and growth form are reported. Growth form categories are based on height classes and include: Lawn (monoculture of turfgrass), Ground Cover (low-growing flowering plants, prostrate or spreading low plants, and creeping or trailing ground covers), Herb (non-graminoid herbaceous plants with vertical growth, including ferns and fern-like plants), Graminoid (grass-like, non-turf species), Shrub (shrub-like evergreen or deciduous plants, including palms and cycads), Subcanopy (tree-like plants and palms up to 10 m), and Canopy (tree-like evergreen or deciduous plants up to 20 m).

Landscape Type	Common Name	Genus	Species	Code	Native Status	Height Class
Del Webb	Christmas Palm	<i>Adonidia</i>	<i>merrillii</i>	ADOMER	N	SubCanopy
	Dogbane gsp. potted	<i>Apocynum</i>	<i>spp.</i>	APOSPP	NA	Herb
	Milkweed	<i>Asclepias</i>	<i>curassavica</i>	ASCCUR	N	Herb
	Foxtail Fern	<i>Asparagus</i>	<i>densiflorus</i>	ASPDEN	N	Herb
	Bougainvillea	<i>Bougainvillea</i>	<i>spp.</i>	BOUSPP	NA	Shrub
	Bromeliads	<i>Bromeliaceae</i>	<i>spp</i>	BROSPP	NA	Herb
	Bottle Brush	<i>Callistemon</i>	<i>citrinus</i>	CALCIT	N	SubCanopy
	Fishtail Palm	<i>Caryota</i>	<i>spp.</i>	CARSPP	N	SubCanopy
	Madagascar periwinkle?	<i>Catharanthus</i>	<i>roseus</i>	CARROS	N	Herb
	Pogota flower	<i>Clerodendrum</i>	<i>paniculatum</i>	CLEPAN	N	Shrub
	Croton	<i>Codiaeum</i>	<i>variegatum</i>	CODVAR	N	Shrub
	Elephant Ear	<i>Colocasia</i>	<i>spp.</i>	COLSPP	N	Herb
	Blue_Mist	<i>Conoclinium</i>	<i>coelestinum</i>	CONCOE	Y	Herb
	Red Cordyline	<i>Cordyline</i>	<i>spp.</i>	CORSPP	NA	Shrub
	Tickseed	<i>Coreopsis</i>	<i>leavenworthii</i>	CORLEA	Y	Herb
	Queen Emma	<i>Crinum</i>	<i>augustum</i>	CRIAUG	N	Herb
	Dianella	<i>Dianella</i>	<i>tasmanica</i>	DIATAS	Y	Herb
	Snake Plant	<i>Dracaena</i>	<i>trifasciata</i>	DRATRI	N	Herb
	Firestar Orchid	<i>Epidendrum</i>	<i>radicans</i>	EPIRAD	Y	Herb
	Knotting ?Spurge	<i>Euphorbia</i>	<i>spp.</i>	EUPSPP	NA	Herb
	Gold Mound Durant	<i>Euphorbia</i>	<i>milii</i>	EUPMIL	N	Shrub
	Blackfoot daisy	<i>Euryops</i>	<i>chrysanthemoides</i>	EURCHR	N	Herb
	Blue Daze	<i>Evolvulus</i>	<i>hybrid</i>	EVOHYB	N	Herb
	Unk legume	<i>Fabaecaeae</i>	<i>spp</i>	FABSPP	NA	Herb
	Blanket flower	<i>Gaillardia</i>	<i>spp.</i>	GAISPP	NA	Herb

Landscape Type	Common Name	Genus	Species	Code	Native Status	Height Class
	Blanketflower	<i>Gaillardia</i>	<i>grandiflora</i>	GAIGRA	N	Herb
	Thryallis	<i>Galphimia</i>	<i>gracilis</i>	GALGRA	N	Shrub
	Firebush	<i>Hamelia</i>	<i>patens</i> 'Calusa'	HAMPAT	Y	Shrub
	Hibiscus	<i>Hibiscus</i>	<i>rosa-sinensis</i>	HIBROS	N	Shrub
	Amaryllis	<i>Hippeastrum</i>	<i>spp.</i>	HIPSPP	N	Herb
	Ixora	<i>Ixora</i>	<i>spp.</i>	IXOSPP	N	Shrub
	Crepe Myrtle	<i>Lagerstroemia</i>	<i>indica</i>	LAGIND	N	SubCanopy
	Sweet Gum	<i>Liquidambar</i>	<i>styraciflua</i>	LIQSTY	Y	Canopy
	Liriope	<i>Liriope</i>	<i>muscari</i>	LIRMUS	N	Herb
	Ryegrass	<i>Lolium</i>	<i>spp.</i>	LOLSPP	N	Graminoid
	Loropetalum	<i>Loropetalum</i>	<i>chinense</i>	LORCHI	N	Shrub
	Phasey_Bean	<i>Macroptilium</i>	<i>lathyroides</i>	MACLAT	N	Herb
	Spearmint	<i>Mentha</i>	<i>spicata</i>	MENSPI	N	Herb
	Bee_Balm	<i>Monarda</i>	<i>punctata</i>	MONPUN	Y	Herb
	Wood sorel spp	<i>Oxalis</i>	<i>spp.</i>	OXASPP	Y	GroundCover
	Dwarf Pentas	<i>Pentas</i>	<i>lanceolata</i> "Dwarf"	PENLAN	Y	Herb
	Congo Rojo Philodendron	<i>Philodendron</i>	<i>tatei</i> 'Congo Rojo'	PHITAT	N	Shrub
	Pygmy Date Palm	<i>Phoenix</i>	<i>roebelenii</i>	PHOROE	N	Shrub
	Plumeria spp (rubra)	<i>Plumeria</i>	<i>rubra</i>	PLURUB	N	SubCanopy
	Podocarpus	<i>Podocarpus</i>	<i>macrophyllus</i>	PODMAC	N	Subcanopy
	Quercus spp.	<i>Quercus</i>	<i>spp.</i>	QUESPP	Y	Canopy
	Live Oak	<i>Quercus</i>	<i>virginiana</i>	QUEVIR	Y	Canopy
	Indian Hawthorn	<i>Raphiolepis</i>	<i>indica</i>	RHAIND	N	Shrub
	UNK011 Rhododendron	<i>Rhododendron</i>	<i>spp.</i>	RHOSPP	N	Shrub
	Florida Snow	<i>Richardia</i>	<i>scabra</i>	RICSCA	N	GroundCover
	Panama Rose (potted)	<i>Rondeletia</i>	<i>leucophylla</i>	RONLEU	N	Shrub
	Schefflera	<i>Schefflera</i>	<i>arboricola</i> 'Trinette'	SCHARB	N	Shrub
	Sedum spp	<i>Sedum</i>	<i>spp.</i>	SEDSPP	NA	Herb
	Silver Saw Palmetto	<i>Serenoa</i>	<i>repens</i> 'Cinera'	SERCIN	Y	Shrub
	St. Augustine Grass	<i>Stenotaphrum</i>	<i>secundatum</i>	STESEC	N	Lawn
	Bird of Paradise	<i>Strelitzia</i>	<i>spp.</i>	STRSPP	N	Herb

Landscape Type	Common Name	Genus	Species	Code	Native Status	Height Class
	White Bird of Paradise	<i>Strelitzia</i>	<i>nicolai</i>	STRNIC	N	Herb
	Asiatic Jasmine	<i>Trachelospermum</i>	<i>asiaticum 'Minima'</i>	TRAMIN	N	GroundCover
	Oyster Plants	<i>Tradescantia</i>	<i>spathacea</i>	TRASPA	N	Herb
	Society Garlic	<i>Tulbaghia</i>	<i>violacea</i>	TULVIO	N	Herb
	Winged elm	<i>Ulmus</i>	<i>alata</i>	ULMALA	Y	Canopy
	Acacia Tree	<i>Vachellia</i>	<i>farnesiana</i>	VACFAR	N	SubCanopy
	Viburnum O	<i>Viburnum</i>	<i>odoratissimum</i>	VIBODO	Y	Shrub
Weslyn Park	Red Maple	<i>Acer</i>	<i>rubrum</i>	ACERUB	Y	Canopy
	Perrenial Peanut	<i>Arachis</i>	<i>glabrata</i>	ARAGLA	N	GroundCover
	Foxtail fern	<i>Asparagus</i>	<i>densiflorus</i>	ASPDEN	N	Herb
	Bidens	<i>Bidens</i>	<i>alba</i>	BIDALB	Y	Herb
	Georgia Calamint	<i>Calamintha</i>	<i>georgiana</i>	CALGEO	Y	Herb
	American Beautyberry	<i>Callicarpa</i>	<i>americana</i>	CALAME	Y	Shrub
	Bottle Brush	<i>Callistemon</i>	<i>citrinus</i>	CALCIT	Y	SubCanopy
	Sedge spp.	<i>Carex</i>	<i>spp.</i>	CARSPP	N	Graminoid
	Tangerine	<i>Citrus</i>	<i>reticulata</i>	CITRET	N	SubCanopy
	Blue Mist	<i>Conoclinium</i>	<i>coelestinum</i>	CONCOE	Y	Herb
	Tickseed	<i>Coreopsis</i>	<i>leavenworthii</i>	CORLEA	Y	Herb
	African Iris	<i>Dietes</i>	<i>iridioides</i>	DIEIRI	N	Herb
	Twin Flower	<i>Dyschoriste</i>	<i>humistrata</i>	DYSHUM	Y	GroundCover
	Lovegrass	<i>Eragrostis elliottii</i>	<i>elliottii</i>	ERAELL	Y	Graminoid
	Orchid	<i>Eulophia</i>	<i>graminea</i>	EULGRA	N	Herb
	Spurge spp.	<i>Euphorbia</i>	<i>spp.</i>	EUPSPP	NA	Herb
	Blanket Flower	<i>Gaillardia</i>	<i>pulchella</i>	GAIPUL	Y	Herb
	Firebush	<i>Hamelia</i>	<i>patens 'Calusa'</i>	HAMPAT	Y	Shrub
	Dollarweed	<i>Hydrocotyle</i>	<i>umbellata</i>	HYDUMB	N	GroundCover
	Dahoon Holly	<i>Ilex</i>	<i>cassine</i>	ILECAS	Y	SubCanopy
	Gallberry	<i>Ilex</i>	<i>glabra</i>	ILEGAL	Y	Shrub
	Yapoun Holly	<i>Ilex</i>	<i>vomitorea</i>	ILEVOM	Y	SubCanopy
	Ocala Anise	<i>Illicium</i>	<i>parviflorum</i>	ILLPAR	Y	Shrub

Landscape Type	Common Name	Genus	Species	Code	Native Status	Height Class
	Jatropha	<i>Jatropha</i>	<i>integerrima</i>	JATINT	N	SubCanopy
	Sweet Gum	<i>Liquidambar</i>	<i>styraciflua</i>	LIQSTY	Y	Canopy
	Loropetalum	<i>Loropetalum</i>	<i>chinense</i>	LORCHI	N	Shrub
	Phasy Bean	<i>Macroptilium</i>	<i>lathyroides</i>	MACLAT	N	Herb
	WD bottlebrush	<i>Melaleuca</i>	<i>viminalis</i>	MELVIM	N	SubCanopy
	Sunshine Mimosa	<i>Mimosa</i>	<i>strigillosa</i>	MIMSTR	Y	GroundCover
	Bee Balm	<i>Monarda</i>	<i>punctata</i>	MONPUN	Y	Herb
	Muhly Grass	<i>Muhlenbergia</i>	<i>capillaris</i>	MUHCAP	Y	Graminoids
	Simpson Stopper	<i>Myrcianthes</i>	<i>fragrans</i>	MYRFRA	Y	SubCanopy
	Wax Myrtle	<i>Myrica</i>	<i>cerifera</i>	MYRCER	Y	Shrub
	Apostles Iris	<i>Neomarica</i>	<i>caerulea 'Regina'</i>	NEOCAE	N	Herb
	Giant Sword Fern	<i>Nephrolepis</i>	<i>biserrata</i>	NEPBIS	Y	Herb
	Royal Fern	<i>Osmunda</i>	<i>regalis</i>	OSMREG	Y	Herb
	Red Rubrum Grass	<i>Pennisetum</i>	<i>setaceum 'rubrum'</i>	PENSET	N	Graminoid
	Penstemon sp.	<i>Penstemon</i>	<i>multiflorus</i>	PENMUL	Y	Herb
	Frog Fruit	<i>Phyla</i>	<i>nodiflora</i>	PHYNOD	Y	GroundCover
	Florida Slash Pine	<i>Pinus</i>	<i>elliotti 'densa'</i>	PINELL	Y	Canopy
	Narrowleaf Silkgrass	<i>Pityopsis</i>	<i>graminifolia</i>	PITGRA	Y	Herb
	Wild Coffee	<i>Psychotria</i>	<i>nervosa</i>	PSYNER	Y	Shrub
	Live Oak	<i>Quercus</i>	<i>virginiana</i>	QUEVIR	Y	Canopy
	Tropical Salvia	<i>Salvia</i>	<i>coccinea</i>	SALCOC	Y	Herb
	Saw Palmetto	<i>Serenoa</i>	<i>repens</i>	SERREP	Y	Shrub
	Silver Saw Palmetto	<i>Serenoa</i>	<i>repens 'Cinera'</i>	SERCIN	Y	Shrub
	Hog Plum	<i>Spondias</i>	<i>mombin</i>	SPOMOM	Y	SubCanopy
	Blue Poterweed	<i>Stachytarpheta</i>	<i>jamaicensis</i>	STAJAM	Y	GroundCover
	Stokes Aster	<i>Stokesia</i>	<i>laevis</i>	STOLAE	Y	Herb
	Bald Cypress	<i>Taxodium</i>	<i>distichum</i>	TAXDIS	Y	Canopy
	Asiatic Jasmine	<i>Trachelospermum</i>	<i>asiaticum 'Minima'</i>	TRAMIN	N	GroundCover
	Fakahatchee Grass	<i>Tripsacum</i>	<i>dactyloides</i>	TRIDAC	Y	Graminoid
	Blueberry Bush	<i>Vaccinium</i>	<i>spp.</i>	VACSPP	Y	Shrub
	Dwarf Walter's Viburnum	<i>Viburnum</i>	<i>obovatum 'Mrs Sc Del'</i>	VIBOBO	Y	Shrub

Landscape Type	Common Name	Genus	Species	Code	Native Status	Height Class
	Cardboard Palm	<i>Zamia</i>	<i>furfuracea</i>	ZAMFUR	N	Shrub
	Coontie Palm	<i>Zamia</i>	<i>integrifolia</i>	ZAMINT	Y	Shrub
	Boston Fern			NEPEXA	Y	Herb

Table S3: Vegetation cover by growth form (height class) in front and back yards at Del Webb (DW) and Weslyn Park (WP) across sampling months. Vegetation cover was estimated using eight cover class categories based on percentage cover: class 0 = not present, class 1 = <1%, class 2 = 1–5%, class 3 = 6–25%, class 4 = 26–50%, class 5 = 51–75%, class 6 = 76–95%, and class 7 = >95%.

Lot Code	Front-Back	Season	GroundCover	Herb	Graminoid	Shrub	SubCanopy	Canopy	Lawn
WP1	B	March	4	2	2	2	3	0	0
WP1	F	March	3	3	0	3	0	0	0
WP2	B	March	3	3	3	4	3	2	0
WP2	F	March	4	0	3	4	2	2	0
WP3	B	March	0	0	3	2	0	2	0
WP3	F	March	3	2	3	3	2	1	0
WP4	B	March	0	0	2	3	3	0	0
WP4	F	March	3	2	2	3	2	0	0
WP5	B	March	3	0	2	2	2	0	0
WP5	F	March	3	1	3	3	3	1	0
WP6	B	March	3	0	2	3	2	0	0
WP6	F	March	0	3	2	3	0	0	0
WP7	B	March	2	3	3	4	2	0	0
WP7	F	March	3	3	3	3	2	0	0
WP8	B	March	3	3	4	2	0	2	0
WP8	F	March	3	3	0	3	2	0	0
WP9	B	March	2	0	3	4	2	2	0
WP9	F	March	2	3	3	4	0	1	0
WP10	B	March	4	3	0	2	3	0	0
WP10	F	March	2	4	3	3	2	3	0
WP11	B	March	0	3	3	3	0	2	0
WP11	F	March	5	0	0	3	3	2	0
WP1	B	June	4	2	3	3	3	2	0
WP1	F	June	3	3	0	3	0	2	0
WP2	B	June	4	3	2	4	2	2	0
WP2	F	June	4	2	2	3	0	3	0
WP3	B	June	0	1	3	3	3	3	0
WP3	F	June	3	2	3	3	3	2	0
WP4	B	June	0	1	2	2	3	0	0
WP4	F	June	4	2	2	4	0	2	0
WP5	B	June	3	0	0	3	2	0	0
WP5	F	June	3	2	0	4	4	2	0
WP6	B	June	4	2	0	3	3	2	0
WP6	F	June	0	3	2	2	0	0	0
WP7	B	June	2	2	5	2	2	3	0
WP7	F	June	3	3	4	3	3	0	0
WP8	B	June	0	3	5	2	0	2	0
WP8	F	June	4	3	3	3	2	0	0
WP9	B	June	2	1	3	3	3	2	0
WP9	F	June	3	3	3	3	3	2	0

Lot Code	Front-Back	Season	GroundCover	Herb	Graminoid	Shrub	SubCanopy	Canopy	Lawn
WP10	B	June	2	5	2	3	0	2	0
WP10	F	June	4	3	0	2	4	0	0
WP11	B	June	0	3	2	3	2	2	0
WP11	F	June	3	0	0	3	3	0	0
WP1	B	September	4	2	3	3	4	2	0
WP1	F	September	4	5	0	3	0	2	0
WP2	B	September	3	3	2	4	2	0	0
WP2	F	September	5	2	3	3	0	3	0
WP3	B	September	0	1	0	3	3	0	0
WP3	F	September	4	2	2	3	2	2	0
WP4	B	September	0	1	2	3	4	0	0
WP4	F	September	4	3	2	3	0	2	0
WP5	B	September	3	0	2	3	3	0	0
WP5	F	September	4	1	3	3	4	1	0
WP6	B	September	4	1	3	3	3	2	0
WP6	F	September	3	4	2	3	0	0	0
WP7	B	September	2	3	4	3	2	2	0
WP7	F	September	3	3	4	4	3	0	0
WP8	B	September	0	4	5	3	0	2	0
WP8	F	September	5	3	4	3	1	0	0
WP9	B	September	2	1	3	3	3	1	0
WP9	F	September	4	1	4	3	2	2	0
WP10	B	September	3	5	2	3	0	2	0
WP10	F	September	5	3	0	2	3	0	0
WP11	B	September	0	4	3	4	1	2	0
WP11	F	September	5	1	0	3	3	0	0
DW1	B	March	2	0	0	2	0	1	5
DW1	F	March	2	3	0	2	2	0	5
DW2	B	March	0	0	0	3	0	0	3
DW2	F	March	0	3	0	2	1	0	5
DW3	B	March	0	1	0	2	3	0	4
DW3	F	March	0	1	0	3	1	0	5
DW4	B	March	0	2	0	3	1	0	4
DW4	F	March	2	3	0	3	1	0	4
DW5	B	March	0	0	0	3	0	1	5
DW5	F	March	0	3	0	3	1	0	4
DW6	B	March	0	2	0	2	1	1	5
DW6	F	March	0	4	0	3	1	0	4
DW7	B	March	0	0	0	0	2	1	5
DW7	F	March	0	3	0	3	0	0	4
DW8	B	March	0	0	0	2	0	1	5
DW8	F	March	0	2	0	3	0	1	5
DW9	B	March	0	2	0	1	0	1	5
DW9	F	March	0	2	0	2	1	0	5

Lot Code	Front-Back	Season	GroundCover	Herb	Graminoid	Shrub	SubCanopy	Canopy	Lawn
DW10	B	March	0	3	0	4	2	0	0
DW10	F	March	0	4	0	3	1	0	4
DW11	B	March	0	0	4	3	0	0	4
DW11	F	March	2	3	0	4	1	0	5
DW12	B	March	0	2	0	3	0	0	5
DW12	F	March	0	2	0	3	2	0	5
DW13	B	March	0	2	0	2	0	0	5
DW13	F	March	3	0	0	3	1	0	5
DW14	B	March	0	3	0	2	0	1	5
DW14	F	March	3	2	0	3	1	0	5
DW1	B	June	2	1	0	2	0	2	5
DW1	F	June	2	3	0	3	3	0	5
DW2	B	June	0	0	0	2	0	0	4
DW2	F	June	0	3	0	3	0	0	5
DW3	B	June	0	2	0	3	0	0	5
DW3	F	June	0	2	0	3	2	0	5
DW4	B	June	0	2	0	4	2	2	5
DW4	F	June	2	3	0	4	2	0	4
DW5	B	June	1	2	0	3	0	2	5
DW5	F	June	0	3	0	3	0	1	5
DW6	B	June	1	2	0	2	1	2	5
DW6	F	June	0	4	0	4	3	0	4
DW7	B	June	0	1	0	0	2	2	5
DW7	F	June	0	2	0	3	0	2	5
DW8	B	June	0	0	0	3	0	2	5
DW8	F	June	0	3	0	3	0	2	5
DW9	B	June	0	2	0	2	2	2	5
DW9	F	June	0	3	0	3	2	0	5
DW10	B	June	0	3	0	2	2	2	4
DW10	F	June	0	4	0	3	3	0	4
DW11	B	June	0	2	0	3	0	0	5
DW11	F	June	2	3	0	3	1	0	5
DW12	B	June	0	4	0	3	0	0	5
DW12	F	June	0	3	0	4	2	0	4
DW13	B	June	0	2	0	2	0	1	5
DW13	F	June	2	0	0	3	2	0	5
DW14	B	June	0	3	0	3	0	2	5
DW14	F	June	4	1	0	3	2	0	4
DW1	B	September	2	0	0	2	0	1	5
DW1	F	September	2	3	0	3	3	0	5
DW2	B	September	0	0	0	3	0	0	5
DW2	F	September	0	4	0	3	0	0	5
DW3	B	September	0	2	0	3	0	0	5
DW3	F	September	0	1	0	4	1	0	5

Lot Code	Front-Back	Season	GroundCover	Herb	Graminoid	Shrub	SubCanopy	Canopy	Lawn
DW4	B	September	0	3	0	5	3	0	4
DW4	F	September	3	3	0	4	2	0	5
DW5	B	September	2	3	0	3	0	1	5
DW5	F	September	0	4	0	3	0	1	5
DW6	B	September	2	2	0	3	2	1	5
DW6	F	September	0	3	0	3	2	0	5
DW7	B	September	0	2	0	0	3	1	5
DW7	F	September	0	2	0	3	0	1	5
DW8	B	September	0	0	0	3	0	1	5
DW8	F	September	0	3	0	3	0	1	5
DW9	B	September	0	3	0	3	1	1	5
DW9	F	September	0	4	0	3	1	0	5
DW10	B	September	0	2	0	2	1	1	4
DW10	F	September	0	4	0	3	2	0	5
DW11	B	September	0	1	0	2	0	0	5
DW11	F	September	2	3	0	3	2	0	5
DW12	B	September	0	2	0	3	0	0	5
DW12	F	September	0	3	0	4	2	0	5
DW13	B	September	0	3	0	3	0	1	5
DW13	F	September	3	0	0	3	2	0	5
DW14	B	September	0	3	0	3	0	1	5
DW14	F	September	4	3	0	3	2	0	4

Table S4: Model selection results for beetle activity density, species richness, and native status. Model selection was conducted using an AIC-based comparison to evaluate generalized linear mixed models assessing the effects of landscape style, vegetation type, and vegetation structure. Trap ID was included as a random effect. The most parsimonious model for each response variable is highlighted in bold, and the corresponding conditional R² value is reported. Model outputs are provided separately for fixed factors and random factors. Signif. codes: 0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1.

Response Variable	Model	AICc	dAICc	df	weight	R2
Activity Density	~ Null	1245.8	23.6	3	<0.001	
	~ Landscape style + Trap ID	1246.4	24.1	4	<0.001	
	~ Month + Trap ID	1247.3	25	5	<0.001	
	~ Landscape style + Month + Trap ID	1248	25.8	5	<0.001	
	~ Landscape style * Month + Trap ID	1222.3	0	8	1	0.48

Response Variable	Factor	Estimate	Std. Error	z value	Pr(> z)
Activity Density	(Intercept)	0.9159	0.2289	4.001	6.32e-05 ***
	Conventional landscape	-0.8841	0.3193	-2.769	0.00563 **
	June	-0.8851	0.2762	-3.204	0.00135 **
	September	-0.7966	0.2724	-2.924	0.00345 **
	Conventional:June	2.0124	0.369	5.453	4.95e-08 ***
	Conventional:September	1.4925	0.3702	4.031	5.54e-05***
Random effects: Trap ID	Variance: 0.8234 Std. Dev.:09074				

Response Variable	Model	AICc	dAICc	df	weight	R2
Activity Density	~ Null	1245.8	39.7	3	<0.001	
	~ Vegetation type + Trap ID	1229.5	23.4	5	<0.001	
	~ Month + Trap ID	1247.3	41.2	5	<0.001	
	~ Vegetation type + Month + Trap ID	1231.6	25.5	7	<0.001	
	~ Vegetation type * Month + Trap ID	1206.1	0	11	1	0.51

Response Variable	Factor	Estimate	Std. Error	z value	Pr(> z)
Activity Density	(Intercept)	0.9331	0.2172	4.296	1.74e-05 ***
	Flower Bed	-1.1519	0.3987	-2.889	0.00386 **
	Turf	-0.7368	0.3519	-2.094	0.03627 *
	June	-0.8527	0.2676	-3.187	0.00144 **

	September	-0.7958	0.2631	-3.025	0.00249 **
	Flower Bed:June	1.0941	0.4786	2.286	0.02225 *
	Turf:June	2.3452	0.402	5.833	5.44e-09 ***
	Flower Bed:September	0.7501	0.4897	1.532	0.12555
	Turf:September	1.8642	0.4042	4.612	3.99e-06 ***
Random effects: Trap ID	Variance: 0714 Std. Dev.: 0.845				

Response Variable	Model	AICc	dAICc	df	weight	R2
Species Richness	~ Null	946.2	0	3	0.42	
	~ Landscape style + Trap ID	947.3	1.1	4	0.25	
	~ Month + Trap ID	947.6	1.4	5	0.21	
	~ Landscape style + Month + Trap ID	948.8	2.6	6	0.12	
	~ Landscape style * Month + Trap ID	NA	NA	8	NA	

Response Variable	Model	AICc	dAICc	df	weight	R2
Species Richness	~ Null	946.2	45.6	3	<0.001	
	~ Vegetation type + Trap ID	931.2	30.6	5	<0.001	
	~ Month + Trap ID	947.6	47	5	<0.001	
	~ Vegetation type + Month + Trap ID	933.1	32.5	7	<0.001	
	~ Vegetation type * Month + Trap ID	900.3	0.0	11	1	0.43

Response Variable	Factor	Estimate	Std. Error	z value	Pr(> z)
Activity Density	(Intercept)	0.4104	0.1492	2.751	0.00595 **
	Flower Bed	-0.9619	0.3055	-3.149	0.00164 **
	Turf	-0.6579	0.2589	-2.541	0.01104 *
	June	-0.6347	0.1956	-3.246	0.00117 **
	September	-0.5725	0.1895	-3.022	0.00251 **
	Flower Bed:June	1.0078	0.3667	2.749	0.00599 **
	Turf:June	1.7344	0.2899	5.983	2.20e-09 ***
	Flower Bed:September	0.6035	0.3839	1.572	0.1159
	Turf:September	1.4425	0.291	4.957	7.17e-07 ***

Random effects: Trap ID	Variance: 0.3352	Std. Dev.: 0.579
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Response Variable	Model	AICc	dAICc	df	weight	R2
Native species	~ Null	403.6	6.8	2	0.0189	
	~ Vegetation type + Trap ID	396.8	0.0	4	0.5726	0.21
	~ Month + Trap ID	406	9.2	4	0.0057	
	~ Vegetation type + Month + Trap ID	398.9	2.2	6	0.1949	
	~ Vegetation type * Month + Trap ID	398.8	2.0	10	0.2079	

Response Variable	Factor	Estimate	Std. Error	z value	Pr(> z)
Native Species	(Intercept)	0.68067	0.23764	2.864	0.00418 **
	Flower Bed	-1.09626	0.39451	-2.779	0.00546 **
	Turf	0.08873	0.35141	0.252	0.80066

Random effects: Trap ID	Variance: 0.6276	Std. Dev.: 0.7922
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Response Variable	Model	AICc	dAICc	df	weight	R2
Non-native Species	~ Null	332.3	40.8	2	<0.001	
	~ Vegetation type + Trap ID	307.1	15.7	4	<0.001	
	~ Month + Trap ID	317.2	25.8	4	<0.001	
	~ Vegetation type + Month + Trap ID	292.8	1.4	6	0.33	
	~ Vegetation type * Month + Trap ID	291.5	0.0	10	0.67	0.40

Response Variable	Factor	Estimate	Std. Error	z value	Pr(> z)
Non-native Species	(Intercept)	-2.34569	0.55214	-4.248	2.15e-05 ***
	Flower Bed	-0.46447	0.93759	-0.495	0.6203
	Turf	0.13522	0.77815	0.174	0.862
	June	0.02132	0.69423	0.031	0.9755
	September	0.40921	0.65274	0.627	0.5307
	Flower Bed:June	1.54755	1.15604	1.339	0.1807
	Turf:June	2.92462	1.0268	2.848	0.0044 **
	Flower Bed:September	0.521	1.15677	0.45	0.6524

	Turf:September	2.08772	0.97511	2.141	0.0323 *
Random effects: Trap ID	Variance: 0.8387	Std. Dev.: 0.9158			

Response Variable	Model	AICc	dAICc	df	weight	R2
Activity Density	~ Null	839.9	15.5	3	<0.001	
	~ Lawn + Trap ID	839	14.6	6	<0.001	
	~ Lawn + Month + Trap ID	840.4	16	8	<0.001	
	~ Lawn * Month + Trap ID	824.4	0	12	1	0.84
	~ GroundCover + Trap ID	847.3	22.9	8	<0.001	
	~ GroundCover + Month + Trap ID	846.8	22.4	10	<0.001	
	~ GroundCover * Month + Trap ID	851	26.6	17	<0.001	
	~ GroundCover + Herb + Graminoid + Trap ID	854.9	30.5	17	<0.001	
	~ (GroundCover + Herb + Graminoid) + Month + Trap ID	855.1	30.7	19	<0.001	
	~ (GroundCover + Herb + Graminoid) * Month + Trap ID	847.9	23.5	42	<0.001	
	~ Shrub + SubCanopy + Canopy + Trap ID	853.7	29.3	15	<0.001	
	~ (Shrub + SubCanopy + Canopy) + Month + Trap ID	852.4	28	17	<0.001	
	~ (Shrub + SubCanopy + Canopy) * Month + Trap ID	876.1	51.7	36	<0.001	

Response Variable	Factor	Estimate	Std. Error	z value	Pr(> z)
Activity Density	(Intercept)	1.5641	0.241	6.489	8.63e-11 ***
	Lawn3	-17.5203	4.431	-0.004	0.996845
	Lawn4	-0.9726	0.4831	-2.013	0.044071 *
	Lawn5	-0.4354	0.3641	-1.196	0.231788
	June	-0.6317	0.289	-2.186	0.028812 *
	September	-0.6409	0.2852	-2.247	0.024614 *
	Lawn3:June	NA	NA	NA	NA
	Lawn4:June	2.3575	0.6103	3.863	0.000112 ***
	Lawn5:June	1.6074	0.4206	3.822	0.000133 ***
	Lawn3:September	NA	NA	NA	NA
	Lawn4:September	2.3732	0.7117	3.335	0.000854 ***
	Lawn5:September	1.1994	0.4159	2.884	0.003932 **
Random effects: Trap ID	Variance: 0.4466	Std. Dev.: 0.6683	Std. Dev.		

Response Variable	Model	AICc	dAICc	df	weight	R2
Species Richness	~ Null	595.3	24.8	3	<0.001	0.88
	~ Lawn + Trap ID	594.4	23.8	6	<0.001	
	~ Lawn + Month + Trap ID	594.1	23.5	8	<0.001	
	~ Lawn * Month + Trap ID	570.6	0.0	12	1	
	~ GroundCover + Trap ID	605	34.5	8	<0.001	
	~ GroundCover + Month + Trap ID	603.5	32.9	10	<0.001	
	~ GroundCover * Month + Trap ID	590	19.4	17	<0.001	
	~ GroundCover + Herb + Graminoid + Trap ID	610.4	39.8	17	<0.001	
	~ (GroundCover + Herb + Graminoid) + Month + Trap ID	604.5	33.9	15	<0.001	
	~ (GroundCover + Herb + Graminoid) * Month + Trap ID	NA	NA	42	NA	
	~ Shrub + SubCanopy + Canopy + Trap ID	604.5	33.9	15	<0.001	
	~ (Shrub + SubCanopy + Canopy) + Month + Trap ID	603.5	32.9	17	<0.001	
	~ (Shrub + SubCanopy + Canopy) * Month + Trap ID	NA	NA	36	NA	

Response Variable	Factor	Estimate	Std. Error	z value	Pr(> z)
Species Richness	(Intercept)	0.9969	0.147	6.782	1.18e-11 ***
	Lawn3	-18.0039	6306.178	-0.003	0.997722
	Lawn4	-0.5981	0.33	-1.813	0.069905 .
	Lawn5	-0.5921	0.2487	-2.381	0.017289 *
	June	-0.528	0.2033	-2.597	0.009412 **
	September	-0.4767	0.2001	-2.383	0.017188 *
	Lawn3:June	NA	NA	NA	NA
	Lawn4:June	1.4721	0.4217	3.491	0.000481 ***
	Lawn5:June	1.4379	0.298	4.826	1.40e-06 ***
	Lawn3:September	NA	NA	NA	NA
	Lawn4:September	1.362	0.4744	2.871	0.004091 **
	Lawn5:September	1.0647	0.2997	3.552	0.000382 ***
Random effects: Trap ID	Variance: 0.134 Std. Dev.: 0.3661	1.0647	0.2997	3.552	0.000382 ***

Response Variable	Model	AICc	dAICc	df	weight	R2
Native Species	~ Null	135.3	0	2	0.6647	

~ Lawn + Trap ID	139.1	3.8	5	0.998
~ Lawn + Month + Trap ID	142.4	7	7	0.0198
~ Lawn * Month + Trap ID	143.8	8.4	11	0.0038
~ GroundCover + Trap ID	138.3	3.0	7	0.149
~ GroundCover + Month + Trap ID	141.2	5.8	9	0.0375
~ GroundCover * Month + Trap ID	147.8	12.5	16	0.0013
~ GroundCover + Herb + Graminoid + Trap ID	142.8	7.5	16	0.016
~ (GroundCover + Herb + Graminoid) + Month + Trap ID	145.6	10.3	18	0.0038
~ (GroundCover + Herb + Graminoid) * Month + Trap ID	NA	NA	41	NA
~ Shrub + SubCanopy + Canopy + Trap ID	152.4	19.1	14	<0.001
~ (Shrub + SubCanopy + Canopy) + Month + Trap ID	158.0	22.7	16	<0.001
~ (Shrub + SubCanopy + Canopy) * Month + Trap ID	NA	NA	35	NA

Response Variable	Model	AICc	dAICc	df	weight	R2
Non-native Species	~ Null	205.9	22.8	2	<0.001	0.58
	~ Lawn + Trap ID	195.7	12.6	5	0.0011	
	~ Lawn + Month + Trap ID	184	1	7	0.3771	
	~ Lawn * Month + Trap ID	183.1	0.0	11	0.6081	
	~ GroundCover + Trap ID	204.8	21.8	7	<0.001	
	~ GroundCover + Month + Trap ID	193.8	10.7	9	0.0028	
	~ GroundCover * Month + Trap ID	191.2	8.2	16	0.0103	
	~ GroundCover + Herb + Graminoid + Trap ID	205.9	22.8	16	<0.001	
	~ (GroundCover + Herb + Graminoid) + Month + Trap ID	197.6	14.5	18	<0.001	
	~ (GroundCover + Herb + Graminoid) * Month + Trap ID	204.9	21.8	41	<0.001	
	~ Shrub + SubCanopy + Canopy + Trap ID	218.1	35	14	<0.001	
	~ (Shrub + SubCanopy + Canopy) + Month + Trap ID	201.4	18.3	16	<0.001	
	~ (Shrub + SubCanopy + Canopy) * Month + Trap ID	222.7	39.7	35	<0.001	

Response	Factor	Estimate	Std. Error	z value	Pr(> z)
Non-native Species	(Intercept)	-1.761E+00	6.283E-01	-2.803	0.00506 **
	Lawn3	-1.664E+01	1.208E+04	-0.001	0.9989
	Lawn4	1.118E+00	1.021E+00	1.096	0.27319

	Lawn5		-5.299E-02	9.080E-01	-0.058	0.95346
	June		3.735E-01	7.825E-01	0.477	0.63319
	September		8.870E-01	7.558E-01	1.174	0.24054
	Lawn3:June		NA	NA	NA	NA
	Lawn4:June		6.999E-01	1.435E+00	0.488	0.62576
	Lawn5:June		3.426E+00	1.279E+00	2.679	0.00738 **
	Lawn3:September		NA	NA	NA	NA
	Lawn4:September		6.814E-01	1.782E+00	0.382	0.70221
	Lawn5:September		1.583E+00	1.123E+00	1.410	0.15853
Random effects: Trap ID	Variance: 0.624	Std. Dev.: 0.7899				

Signif. codes: 0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Table S5: Estimated marginal means (EMMs) on the response (probability) scale and Tukey-adjusted pairwise comparisons among vegetation types (Native, Flower Bed, Turf) for native and non-native responses. Odds ratios, standard errors (SE), z-ratios, and adjusted p-values are shown for pairwise contrasts. Estimated probabilities with asymptotic 95% confidence intervals (CI) are reported for each vegetation type.

Native species						
contrast	odds ratio	SE	df	null	z.ratio	p.value
Native / FlowerBed	2.993	1.18	Inf	1	2.779	0.0151
Native / Turf	0.915	0.322	Inf	1	-0.252	0.9655
FlowerBed / Turf	0.306	0.127	Inf	1	-2.855	0.012
Vegetation type	prob	SE	df	asympt.LCL	asympt.UCL	
Native	0.664	0.053	Inf	0.554	0.759	
FlowerBed	0.398	0.0733	Inf	0.266	0.546	
Turf	0.683	0.0592	Inf	0.558	0.787	
Non-native species						
contrast	odds ratio	SE	df	null	z.ratio	p.value
Month = March:						
Native / FlowerBed	1.5912	1.49	Inf	1	0.495	0.8735
Native / Turf	0.8735	0.68	Inf	1	-0.174	0.9835
FlowerBed / Turf	0.549	0.536	Inf	1	-0.614	0.8126
Month = June:						
Native / FlowerBed	0.3385	0.256	Inf	1	-1.435	0.323
Native / Turf	0.0469	0.0341	Inf	1	-4.204	0.0001
FlowerBed / Turf	0.1385	0.101	Inf	1	-2.702	0.0189
Month = September:						
Native / FlowerBed	0.945	0.72	Inf	1	-0.074	0.997
Native / Turf	0.1083	0.0714	Inf	1	-3.371	0.0022
FlowerBed / Turf	0.1146	0.0887	Inf	1	-2.798	0.0142
Vegetation type	prob	SE	df	asympt.LCL	asympt.UCL	
Month = March:						
Native	0.0874	0.044	Inf	0.0314	0.22	
FlowerBed	0.0568	0.0442	Inf	0.0118	0.233	
Turf	0.0988	0.0555	Inf	0.0313	0.271	
Month = June:						
Native	0.0891	0.0449	Inf	0.032	0.244	
FlowerBed	0.2242	0.0989	Inf	0.0866	0.468	
Turf	0.676	0.0947	Inf	0.4719	0.83	
Month = September:						
Native	0.126	0.0545	Inf	0.0519	0.275	
FlowerBed	0.1324	0.0735	Inf	0.0417	0.349	
Turf	0.5711	0.104	Inf	0.3665	0.754	

Table S6: Estimated marginal means (EMMs) on the response (probability) scale and Tukey-adjusted pairwise comparisons among lawn cover for non-native beetle response. Odds ratios, standard errors (SE), z-ratios, and adjusted p-values are shown for pairwise contrasts. Estimated probabilities with asymptotic 95% confidence intervals (CI) are reported for each vegetation type.

Non-native species							
contrast	odds ratio	SE	df	null	z.ratio	p.value	
Month = March:							
Lawn0 / Lawn3	16926313	2.04e+11	Inf		1	0.001	1.0000
Lawn0 / Lawn4	0	0.00e+00	Inf		1	-1.096	0.6921
Lawn0 / Lawn5	1	1.00e+00	Inf		1	0.058	0.9999
Lawn3 / Lawn4	0	0.00e+00	Inf		1	-0.001	1.0000
Lawn3 / Lawn5	0	0.00e+00	Inf		1	-0.001	1.0000
Lawn4 / Lawn5	3	3.00e+00	Inf		1	1.091	0.6951
Month = June:							
Lawn0 / Lawn3	nonEst	NA	NA		1	NA	NA
Lawn0 / Lawn4	0	0 0.00e+00	Inf		1	-1.678	0.2134
Lawn0 / Lawn5	0	0 0.00e+00	Inf		1	-3.56	0.0011
Lawn3 / Lawn4	nonEst	NA	NA		1	NA	NA
Lawn3 / Lawn5	nonEst	NA	NA		1	NA	NA
Lawn4 / Lawn5	0	0 0.00e+00	Inf		1	-1.414	0.3335
Month = September:							
Lawn0 / Lawn3	nonEst	NA	NA		1	NA	NA
Lawn0 / Lawn4	0	0 0.00e+00	Inf		1	-1.193	0.4573
Lawn0 / Lawn5	0	0 0.00e+00	Inf		1	-2.104	0.089
Lawn3 / Lawn4	nonEst	NA	NA		1	NA	NA
Lawn3 / Lawn5	nonEst	NA	NA		1	NA	NA
Lawn4 / Lawn5	1	2.00e+00	Inf		1	0.185	0.9814

Lawn cover	prob	SE	df	asympt.LCL	asympt.UCL
Month = March:					
0	0.147	0.0786	Inf	0.0478	0.371
3	0.000	0.000123	Inf	0.000	1.000
4	0.345	0.189	Inf	0.0927	0.73
5	0.14	0.0867	Inf	0.0382	0.401
Month = June:					
0	0.2	0.0933	Inf	0.0737	0.439
3	nonEst	NA	NA	NA	NA
4	0.606	0.212	Inf	0.2131	0.897
5	0.879	0.0743	Inf	0.6487	0.966
Month = September:					
0	0.294	0.109	Inf	0.1294	0.54
3	nonEst	NA	NA	NA	NA
4	0.716	0.284	Inf	0.1407	0.975
5	0.658	0.108	Inf	0.4283	0.832

Table S7: Indicator species analysis (IndVal.g) identifying species significantly associated with vegetation groups. The analysis was conducted using a significance level of $\alpha = 0.05$ across 74 taxa. Indicator value components A (specificity) and B (fidelity), the IndVal.g statistic, and permutation-based p-values are reported. Significance levels are 0 '****' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1.

Group Turf #sps. 8

	A	B	stat	p.value
<i>Gondwanocrypticus platensis</i>	0.83721	0.23711	0.446	0.001 ***
<i>Heteroderes amplicollis</i>	0.60888	0.2268	0.372	0.004 **
<i>Ahasverus rectus</i>	0.77623	0.1134	0.297	0.018 *
<i>Vacusus parvus</i>	0.78775	0.09278	0.270	0.005 **
<i>Xantholininae sp.A</i>	1.00000	0.05155	0.227	0.006 **
<i>Acanthinus scitulus</i>	0.89015	0.05155	0.214	0.033 *
<i>Lithocharodes ruficollis</i>	0.81664	0.05155	0.205	0.032 *
<i>Euconnus sp.A</i>	1.00000	0.03093	0.176	0.038 *

Group Native+Turf #sps. 1

	A	B	stat	p.value
<i>Rismethus squamiger</i>	0.9278	0.2237	0.456	0.009 **

Group FlowerBed+Turf #sps. 1

	A	B	stat	p.value
<i>Tetragonoderus laevigatus</i>	0.98351	0.08284	0.285	0.011 *

Table S8: Functional entities (FEs) of beetle taxa recorded in Del Webb (DW) and Weslyn Park (WP) landscapes. For each functional entity, landscape occurrence, total number of taxa, relative percentage contribution, and associated functional traits are shown, including diet, wing morphology, body size, and egg-larval substrate. See Table S1 for categories abbreviations.

FE	Landscape	Total taxa	Percentage	Diet	Wings	Size	Substrate
FE_1	DW-WP	15	20	PRD	MAC	S	DOM
FE_2	DW-WP	9	12	DET	MAC	S	DOM
FE_3	DW-WP	8	10.67	MYC	MAC	S	DOM
FE_4	DW-WP	5	6.67	OMN	MAC	S	DOM
FE_5	DW-WP	4	5.33	MYC	MAC	S	FNG
FE_6	DW-WP	4	5.33	MYC	BRA	S	DOM
FE_7	DW-WP	4	5.33	PRD	BRA	S	DOM
FE_8	DW-WP	4	5.33	PHY	MAC	S	LPL
FE_9	DW-WP	3	4	PHY	MAC	M	SOI
FE_10	DW-WP	3	4	OMN	MAC	S	SOI
FE_11	DW-WP	2	2.67	PRD	MAC	S	LPL
FE_12	DW-WP	2	2.67	PRD	MAC	S	SOI
FE_13	DW-WP	2	2.67	PHY	MAC	S	DOM
FE_14	DW	1	1.33	MYC	MAC	S	SOI
FE_15	DW-WP	1	1.33	DET	BRA	S	SOI
FE_16	WP	1	1.33	DET	MAC	S	SOI
FE_17	DW	1	1.33	OMN	MAC	M	SOI
FE_18	WP	1	1.33	DET	MAC	M	DOM
FE_19	WP	1	1.33	TRO	MAC	S	SOI
FE_20	WP	1	1.33	PRD	BRA	L	SOI
FE_21	DW	1	1.33	PRD	MAC	M	DOM
FE_22	DW	1	1.33	PRD	BRA	S	SOI

Table S9: Functional traits of beetles, with species counts (n) and percentages (%) within Del Webb (conventional landscape) and Weslyn Park (native-dominated landscape), classified by native status (Y = native, N = non-native). Traits include diet, wing morphology, body size, and egg-larval substrate preference. Diet trait states are OMN (omnivorous), PRD (predatory), PHY (phytophagous), MYC (mycophagous), DET (detritivorous), and TRO (trophallaxis). Wing trait states are BRA (brachypterous) and MAC (macropterous). Body size classes are S (small), M (medium), and L (large). Egg-larval substrate trait states are SOI (soil), DOM (decayed organic matter), LPL (live plant), and FNG (fungus).

Site	Trait	Trait State	Native Status	Species (n)	% Within Site
Del Webb	Diet	DET	Y	7	14
			N	2	4
		MYC	Y	9	18
			N	2	4
		OMN	Y	5	10
			N	3	6
		PHY	Y	6	12
			N	0	0
		PRD	Y	15	30
			N	1	2
Weslyn Park	Diet	DET	Y	8	16.3
			N	1	2
		MYC	Y	12	24.5
			N	2	4.1
		OMN	Y	2	4.1
			N	3	6.1
		PHY	Y	6	12.2
			N	0	0
		PRD	Y	13	26.5
			N	1	2
Del Webb	Substrate	DOM	Y	28	56
			N	5	10
		FNG	Y	2	4
			N	0	0
		LPL	Y	3	6
			N	1	2
		SOI	Y	9	18
			N	2	4
Weslyn Park	Substrate	DOM	Y	26	53.1
			N	4	8.2
		FNG	Y	4	8.2
			N	0	0
		LPL	Y	3	6.1
			N	1	2
		SOI	Y	8	16.3
			N	3	6.1
Del Webb	Wings	BRA	Y	7	14
			N	N	2
		MAC	Y	Y	70
			N	N	14
Weslyn Park	Wings	BRA	Y	7	14.3
			N	1	2
		MAC	Y	34	69.4
			N	7	14.3

Site	Trait	Trait State	Native Status	Species (n)	% Within Site
Del Webb	Size	L	Y	0	0
			N	0	0
		M	Y	Y	10
			N	N	0
		S	Y	Y	74
			N	N	16
Weslyn Park	Size	L	Y	1	2
			N	0	0
		M	Y	4	8.2
			N	0	0
		S	Y	36	73.5
			N	8	16.3